

PERT

PERT

- Stands for **P**rogram **E**valuation and **R**everview **T**echnique.
- **PERT** is developed to take account of the uncertainty surrounding estimates of task durations.
- It is very similar to the CPM technique but instead of using one estimate for the duration of each task, it uses 3 estimates.
- Helps estimate the probability of not meeting or missing the target date
- represents the outcome of future events, based on very limited information.
- Unlike CPM approach which specifies “the completion date for the project is”...., PERT specifies, “we expect to complete the project by....”.

- PERT uses **three estimates** unlike one estimate used earlier.
 - **Optimistic (*a*) duration of an activity** : the shortest time that we expect to complete the activity,
 - **Most likely (*m*) duration of an activity**: the time we expect the task to take under normal circumstances,
 - **Pessimistic (*b*) duration of an activity**: the worst possible time.
 - **PERT** combines these three duration to form a single **Expected**
 - **Duration** (t_e) = $a + 4m + b / 6$
 - **The expected duration (t_e)** is used to carry out the forward pass through the network.

Using PERT to evaluate the effects of uncertainty

Three estimates are produced for each activity

- *Most likely time (m)*
- *Optimistic time (a)*
- *Pessimistic (b)*
- ‘expected time/duration’ $t_e = (a + 4m + b) / 6$

In this approach, the event dates obtained are **not the earliest possible dates** but the **dates by which we expect to achieve those events.**

The earliest time and variance of each activity is computed by using the formula.

$$t_e = \frac{a + 4m + b}{6} \text{ and } \sigma^2 = \left(\frac{b - a}{6} \right)^2$$

Variance project length is obtained by summing variance of each critical activity.

The expected project length is the sum of duration of each critical activity.

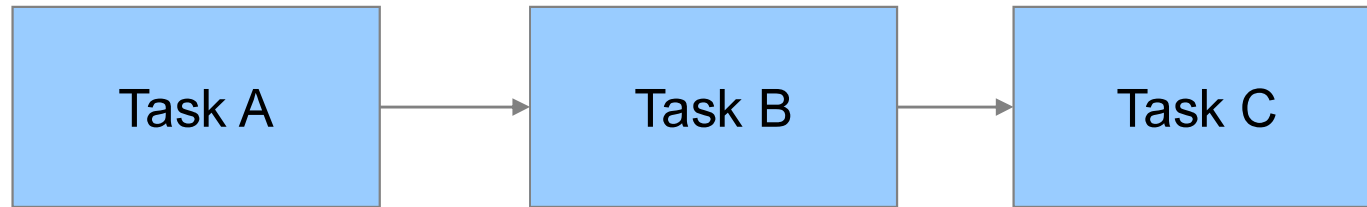
Activity standard deviation: S

- A quantitative measure of the **degree of uncertainty of an activity during estimate**
- $S = (b-a)/6$
- S (Standard Deviation) is **proportional to the difference between optimistic and pessimistic estimates to rank the degree of uncertainty or risk for each activity.**

Steps followed

- Determine the expected duration
- Calculate the standard deviation for each project event
- Calculate the z value for each event that has a target date
- Convert z values to a probability.

A chain of activities



Task	a	m	b	t_e	s
A	10	12	16	?	?
B	8	10	14	?	?
C	20	24	38	?	?

Expected duration (t_e) & Standard Deviation(s)

Task A: $t_e = (10 + (12 \times 4) + 16) / 6$ i.e. 12.33

$$s = (16 - 10) / 6 \text{ i.e. } 1$$

Task B: $t_e = (8 + (10 \times 4) + 14) / 6$ i.e. 10.33

$$s = (14 - 8) / 6 \text{ i.e. } 1$$

Task C: $t_e = (20 + (24 \times 4) + 38) / 6$ i.e. 25.66

$$s = (38 - 20) / 6 \text{ i.e. } 3$$

A chain of activities (all 3 tasks)

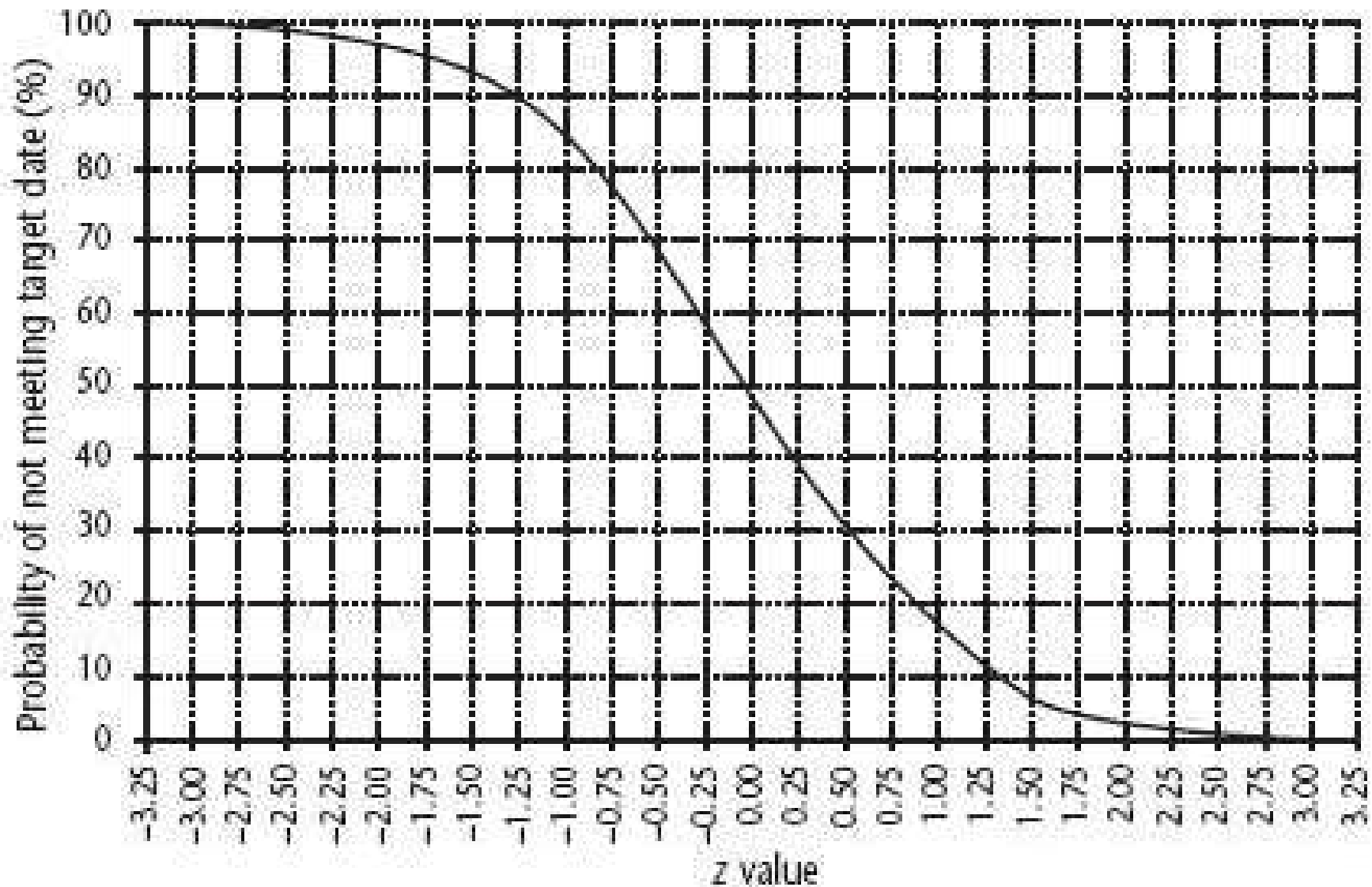
- What would be the **expected duration** of the chain $A + B + C$?
- Answer: $12.33 + 10.33 + 25.66$ i.e. 48.32
- What would be the **standard deviation** for $A + B + C$?
- Answer: square root of $(1^2 + 1^2 + 3^2)$ i.e. 3.32

Assessing the likelihood of meeting a target

- Say the **target for completing A+B+C** was 52 days (T)

Calculate the z value,

- (z) is the difference between **the desired (Target) completion time and the project's expected time** divided by the **standard deviation** for the project.
- The Z-value is calculated for **each node** that has a target date.
- thus $z = (T - t_e)/s$
- Hence, $z = (52-48.32)/3.32$ i.e. 1.1



Graph of z values (Look up in table of z values)
(Probability of not meeting target date)

There is only **15% chance of not meeting** the target of 52 days.

Activity	Duration (weeks)	Precedents
A Hardware selection	6	
B System configuration	4	
C Install hardware	3	A
D Data migration	4	B
E Draft office procedures	3	B
F Recruit staff	10	
G User training	3	E, F
H Install and test system	2	C, D

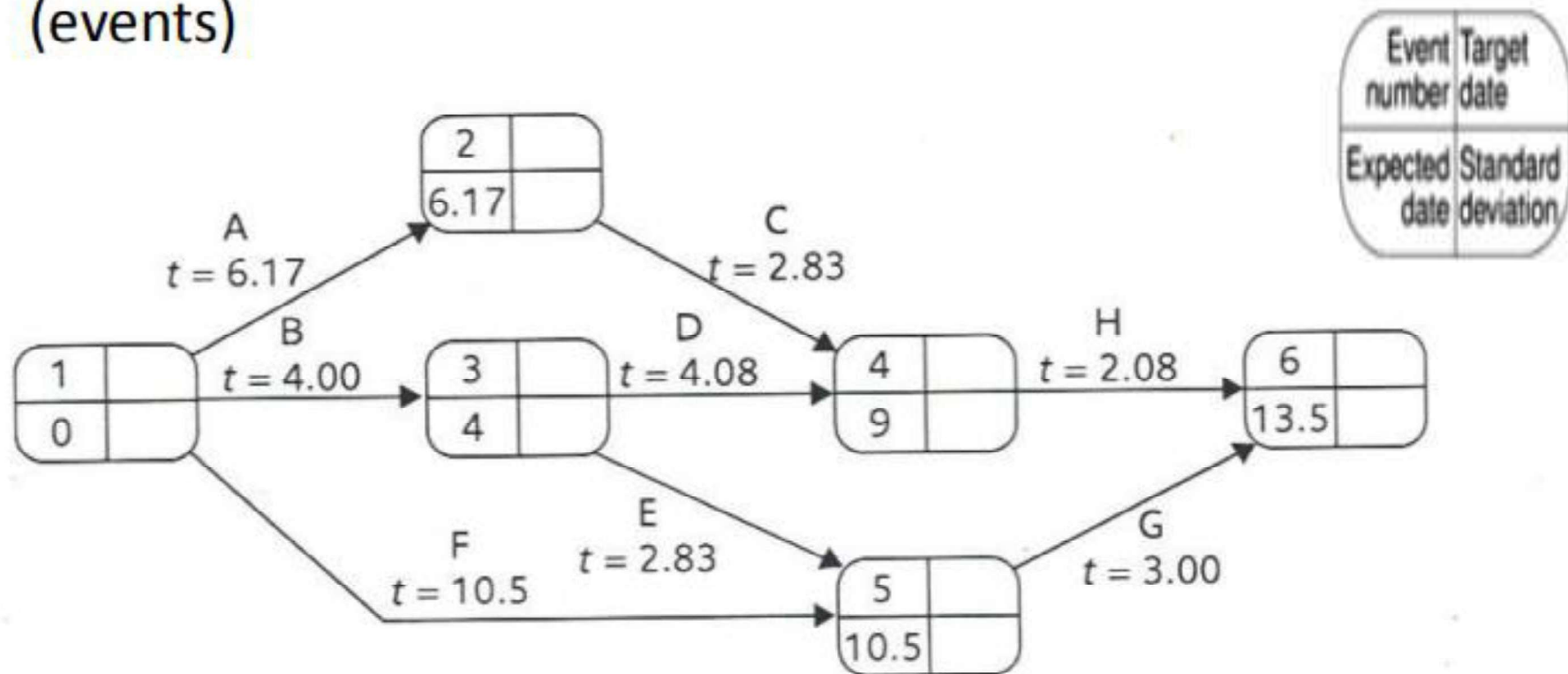
Table 7.3 *PERT activity time estimates*

<i>Activity</i>	<i>Activity durations (weeks)</i>		
	<i>Optimistic (a)</i>	<i>Most likely (m)</i>	<i>Pessimistic (b)</i>
A	5	6	8
B	3	4	5
C	2	3	3
D	3.5	4	5
E	1	3	4
F	8	10	15
G	2	3	4
H	2	2	2.5

Table 7.4 *Expected times and standard deviations*

<i>Activity</i>	<i>Activity durations (weeks)</i>				
	<i>Optimistic (a)</i>	<i>Most likely (m)</i>	<i>Pessimistic (b)</i>	<i>Expected (t_e)</i>	<i>Standard deviation (s)</i>
A	5	6	8	6.17	0.50
B	3	4	5	4.00	0.33
C	2	3	3	2.83	0.17
D	3.5	4	5	4.08	0.25
E	1	3	4	2.83	0.50
F	8	10	15	10.50	1.17
G	2	3	4	3.00	0.33
H	2	2	2.5	2.08	0.08

- The PERT Network After using the expected task durations to carry the Forward Pass and Calculating the Expected Date (events)



What does it mean that the project duration is 13.5 weeks.

This means that we expect that the project will be completed halfway through week 14 but because it is an expected value it could finish earlier or later but most likely will take 13.5 weeks.

- **Expected duration t_e provides the expected dates (not earliest date) by which the events would be achieved/completed.**
- **The project is expected to take 13.5 weeks.**
- **This emphasizes the uncertainty of real world. So, we take approximates.**

The PERT technique uses the following three-step method for calculating the probability of meeting or missing a target date:

1. Calculate the standard deviation of each project event.
2. Calculate the z value for each event that has a target value.
3. Convert z values to probabilities.

Standard Deviation for each project event:

- It can be calculated by carrying a forward pass using the activity standard deviations.
- Adding two standard deviations = add their squares and then find the square root of the result
- The standard deviation of the first event = 0
- If an event depends on only one activity, the standard deviation of the event is the same as the standard deviation of the activity.
- If there is more than one path we select the highest standard deviation.

- SD of the event 2 depends on activity A. Therefore the SD of event 2 is 0.5.
- SD of the event 3 depends on activity B. Therefore the SD of event 3 is 0.33.
- SD of the event 5 depends on activity F. Therefore the SD of event 5 is 1.17.

- SD for event 4 depends on two possible paths:
AC or BD.
- Therefore, SD of A+C = $\text{SQRT}(0.5^2 + 0.17^2) = 0.528 = 0.53$
- SD of B+D = $\text{SQRT}(0.33^2 + 0.25^2) = 0.414$
- Therefore, SD for event 4 is **0.53 which is greater than 0.414.**

- SD for event 5 depends on two possible paths:
B+E or F.
- Therefore, SD of B+E = $\text{SQRT}(0.33^2 + 0.50^2) = 0.599 = 0.6$
- SD of F = 1.17
- Therefore SD for event 5 is **1.17 which is greater than 0.6.**

- SD for event 6 depends on two possible paths:
H and G.
- Therefore, SD of H = $\text{SQRT}(0.53^2 + 0.08^2) = 0.53$
- SD of G = $\text{SQRT}(1.17^2 + 0.33^2) = 1.22$
- Therefore, SD for event 6 is **1.22 which is greater than 0.53**

Calculating Z Value

$$Z = \frac{T - T_e}{S}$$

T – Target date, T_e is expected date, S is SD.

Calculate the z values for the event number 4, 5 and 6 where the target date are 10, 10 and 15 respectively. Find the probabilities that those events will not achieve their target dates using the given Z table .

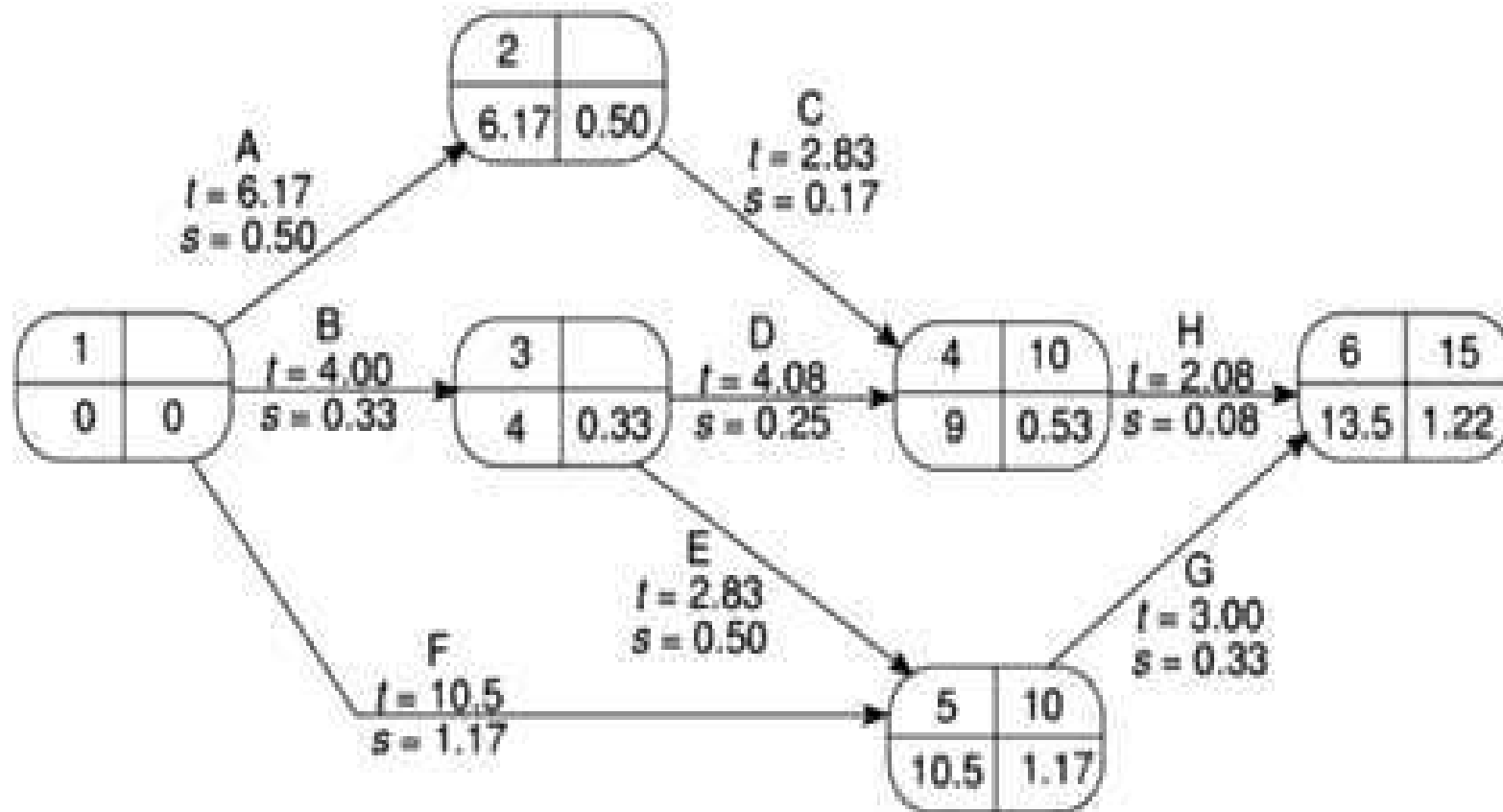
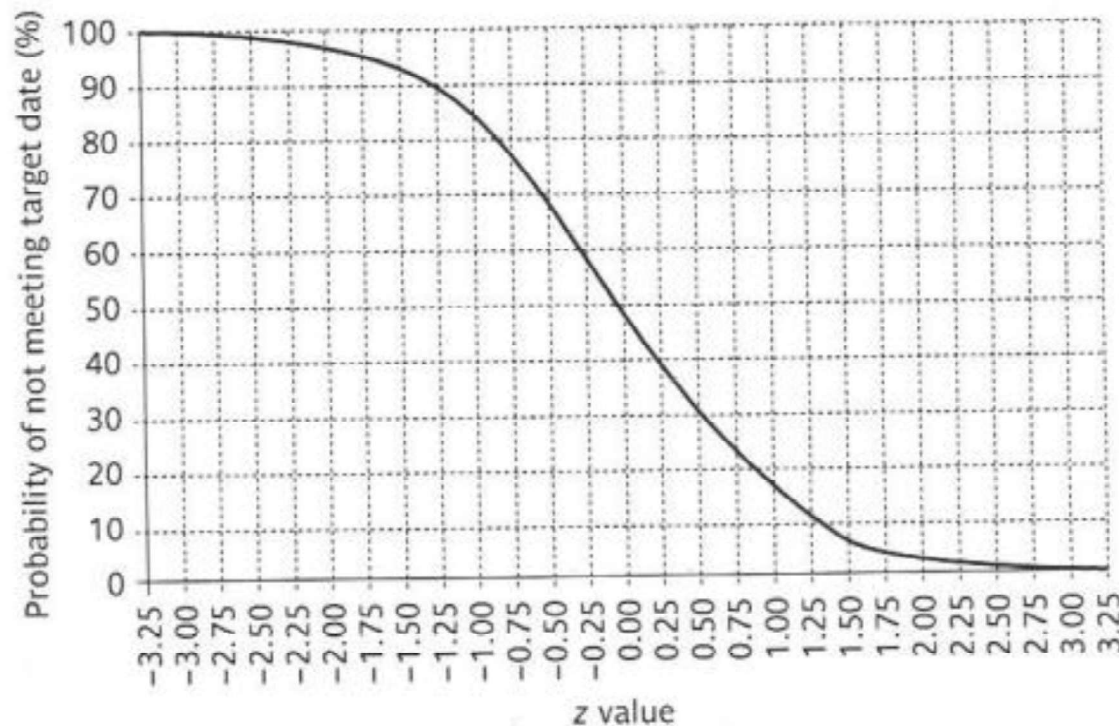


Figure 7.4 *The PERT network with three target dates and calculated event standard deviations.*

Calculating the (z) values:

- The z value is calculated for each node that has a target date.
- This means we can calculate it for events(4, 5, and 6).
- $Z = T(\text{target date}) - t_e (\text{expected date}) / s$
- For event 4:
- $Z = (10 - 9) / 0.53 = 1.89$
- For event 5:
- $Z = (10 - 10.5) / 1.17 = -0.43$
- For event 6:
- $Z = (15 - 13.5) / 1.22 = 1.23$

Graph for Converting z values into Probabilities



The z value can be converted into the probability of NOT meeting the target date by using the figure above.

Z Table

z	x.x0	x.x1	x.x2	x.x3	x.x4	x.x5	x.x6	x.x7	x.x8	x.x9
3	0.00135									
2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.002	0.0019
2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.003	0.0029	0.0028	0.0027	0.0026
2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.004	0.0039	0.0038	0.0037	0.0036
2.5	0.0062	0.006	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
2.4	0.0082	0.008	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.011
2.1	0.0179	0.0174	0.017	0.0166	0.0162	0.0158	0.0154	0.015	0.0146	0.0143
2	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.025	0.0244	0.0239	0.0233
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.5	0.0668	0.0655	0.0643	0.063	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.102	0.1003	0.0985
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.123	0.121	0.119	0.117
1	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.166	0.1635	0.1611
0.8	0.2119	0.209	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.7	0.242	0.2389	0.2358	0.2327	0.2297	0.2266	0.2236	0.2206	0.2177	0.2148
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.5	0.3085	0.305	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.281	0.2776
0.4	0.3446	0.3409	0.3372	0.3336	0.33	0.3264	0.3228	0.3192	0.3156	0.3121
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.352	0.3483
0.2	0.4207	0.4168	0.4129	0.409	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0	0.5	0.496	0.492	0.488	0.484	0.4801	0.4761	0.4721	0.4681	0.4641

For event 4:

$$Z = (10 - 9) / 0.53 = 1.89$$

• **For event 5:**

$$Z = (10 - 10.5) / 1.17 = -0.43$$

• **For event 6:**

$$z = (15 - 13.5) / 1.22 = 1.23$$

The z value for event 4 is 1.89 which equates to the probability of approximately 3%.

The z value for event 5 is -0.43 which equates to the probability of approximately 67%.

The z value for event 6 is 1.23 which equates to the probability of approximately 11%.

A small project consisting of ten activities has the following characteristics:

<i>Activity</i>	<i>Preceding Activity</i>	<i>Time Estimate weeks</i>		
		<i>Optimistic</i>	<i>Most likely</i>	<i>Pessimistic</i>
A	—	4	5	12
B	—	1	1.5	5
C	A	2	3	4
D	A	3	4	11
E	A	2	3	4
F	C	1.5	2	2.5
G	D	1.5	3	4.5
H	B,E	2.5	3.5	7.5
I	H	1.5	2	2.5
J	F,G,I	1	2	3

Perform the following

1. Draw the PERT network for the given project
2. Calculate the expected duration, variance and standard deviation of the project.
3. Prepare the activity schedule for the project and identify the critical path.